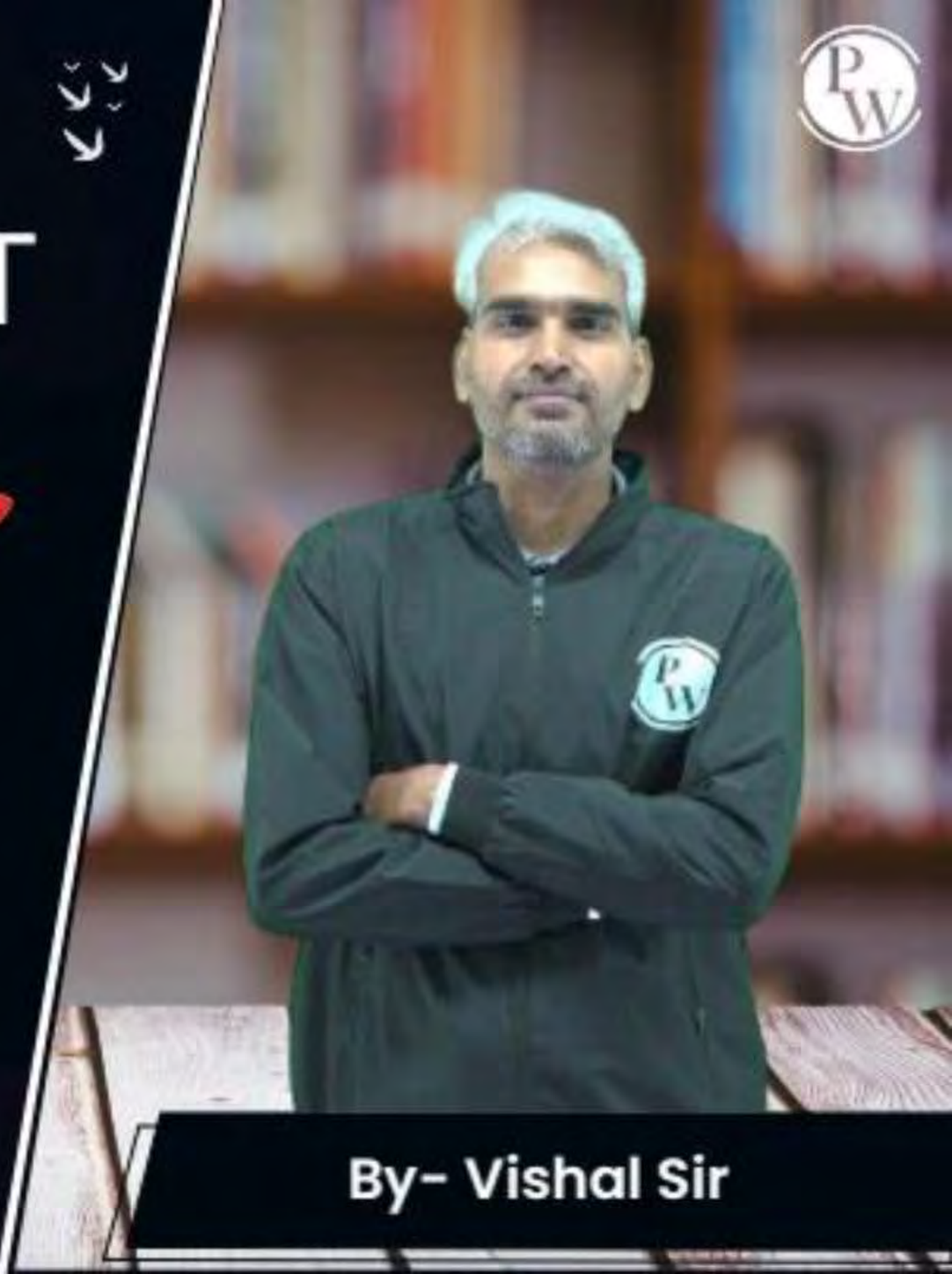
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Number of subsets of a set

Topic Power set

Slide

Topics to be Covered



- Topic — Venn diagram
- Topic — Set operations
- Topic — Properties of set operations

Slide



Topic : Power Set

Power Set: Let 'A' is any finite set, then power set of set 'A' is a set of all subsets of set A.

Power set of set A is denoted by $P(A)$ or 2^A

$P(A)$ = Set of all subsets of set A

$P(A) = \{$ ... $\}$

{ : it is a set

... : Elements will be all subsets of set A

Slide

Q: Let $A = \{a, b, c\}$

then

$P(A) = ?$

Soln.

$$P(A) = \{\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

Note:- ① $P(A)$ is also a set, and we know that $\emptyset$ is a subset of every set

$\therefore \emptyset \subseteq P(A)$ is true for any set A.

② $\emptyset$ is always an element of $P(A)$, for any set A.

i.e. $\emptyset \in P(A)$ is true for any set A.



Topic : Cardinality of Power Set

Cardinality of $P(A)$ = $|P(A)|$ = No. of elements in $P(A)$

= Total no. of subsets of set A.

$$= 2^{|A|}$$

$$\boxed{|P(A)| = 2^{|A|}}$$

Slide

Q:. Let $\emptyset$ be an Empty set. then

$$|P(P(\emptyset))| = ?$$

(a) 0

(b) 1

(c) 2

(d) Can not be defined.

We know, $|\emptyset| = 0$

$$|P(A)| = 2^{|A|} \quad \therefore |P(\emptyset)| = 2^{|\emptyset|} = 2^{0} = 2 = 1$$

Hence,

$$|P(P(\emptyset))| = 2^{|P(\emptyset)|} = 2^{1} = 2$$

Q: Let $\emptyset$ be an Empty set, then

$$|P(P(\emptyset))| = ?$$

(a) 0
(b) 1
(c) 2
(d) Can not be defined.

We must understand

The only subset of $\emptyset$ is $\emptyset$ itself.

$P(\emptyset)$ = Set of all subsets of $\emptyset$

$= \{ \emptyset \}$ ($\emptyset$: subset of $\emptyset$)

$P(P(\emptyset))$ = Set of all subsets of $P(\emptyset)$, i.e. Set of all subsets of $\{\emptyset\}$

$= \{ \{\}, \{\emptyset\} \}$

There are the only two subsets of $\{\emptyset\}$

$$|\{\emptyset, \{\emptyset\}\}| = 2$$

$$\emptyset \neq \{\emptyset\}$$

$\{\}$ it is an Empty set

$\neq$

$\{\{\}\}$ it is a set Containing one element. And that element is an Empty set

$$|\{\}| = 0$$

$$|\{\{\}\}| = 1 = |\{\emptyset\}|$$

Q: Let $A = \{1, 2, 3, \{1,2\}, \{\{2,3\}\}\}$

Which of the following is/are true

{Exactly How many of the given options are True = 11 = Ans}

(a) $1 \in A$ ✓

(b) $\{1\} \in A$ ✗

(c) $1 \subseteq A$ ✗

(d) $\{1\} \subseteq A$ ✓

it is a member of A
∴ True

(e) $\{1,2\} \in A$ ✓

(f) $\{1,2\} \subseteq A$ ✓

all are elements of set A ∴ True

(g) $\{\{1,2\}\} \in A$ ✗

(h) $\{\{1,2\}\} \subseteq A$ ✓

(i) $\{2,3\} \in A$ ✗

(j) $\{2,3\} \subseteq A$ ✓

(k) $\{\{2,3\}\} \in A$ ✓

(l) $\{\{2,3\}\} \subseteq A$ ✗

(m) $\{\{\{2,3\}\}\} \subseteq A$ ✓

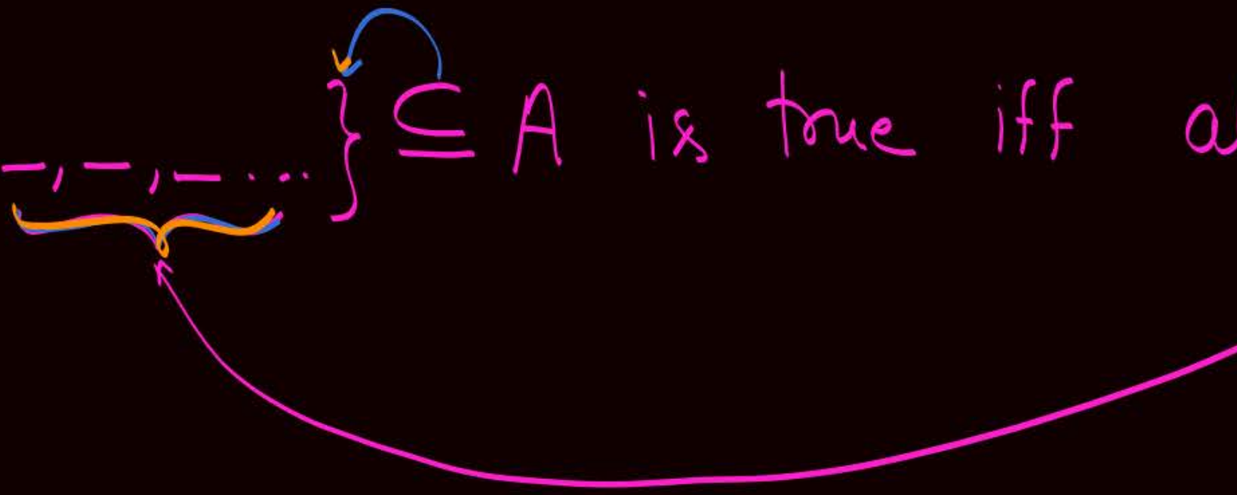
(n) $\emptyset \in A$ ✗

(o) $\emptyset \in P(A)$ ✓

(x) $\emptyset \subseteq A$ ✓

(z) $\emptyset \subseteq P(A)$ ✓

- $a \in A$ is true iff 'a' is a member of set A.
- $\{_, _, _ \ldots\} \subseteq A$ is true iff all elements are member of set A.



Today's Topic

H.W.

Q1. For a set A, the power set of A is denoted by 2^A. If $A = \{5, \{6\}, \{7\}\}$ which of the following options are true?

$P(A) = 2^A = \{\{\}, \{5\}, \{\{6\}\}, \{\{7\}\}, \{5, \{6\}\}, \{5, \{7\}\}, \{\{6\}, \{7\}\}, \{5, \{6\}, \{7\}\}\}$

A. $\emptyset \in 2^A$

B. $\emptyset \subseteq 2^A$ $\equiv \{\} \subseteq 2^A$

C. $\{5, \{6\}\} \in 2^A$

D. $\{5, \{6\}\} \subseteq 2^A$ ✗ (X)

Slide

H.W: GATE-PYQ

Q2. Let $P(S)$ denote the power set of a set S. Which of the following is always true?

it can be any set

A. $P(P(S)) = P(S)$ ⇒ $|P(S)| = 2^{|S|}$ & $|P(P(S))| = 2^{|P(S)|} = 2^{(2^{|S|})}$

$2^{|S|} \neq 2^{(2^{|S|})}$ ∴ $P(S) \neq P(P(S))$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

E. None of the above

Slide

H.W: GATE-PYQ

Q2. Let $P(S)$ denote the power set of a set S. Which of the following is always true?

it can be any set

A. $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

larger set, smaller set

D. $S \notin P(S)$ — $S \in P(S)$ is always True

E. None of the above

$S = \{1\}$

$P(S) = \{\emptyset, \{1\}\}$

$P(P(S)) = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \{\emptyset, \{1\}\}\}$

$P(S) \cap P(P(S)) = \{\emptyset\}$

↳ option B seems Correct. but it need not be true for all set 'S'.

eg: Let $S = \{1, \{1\}\}$

$P(S) \cap P(P(S)) = \{\emptyset, \{\{1\}\}\} \neq \{\emptyset\}$

Slide

let $S = \emptyset$

$P(S) = \{ \emptyset \}$

$P(P(S)) = \{ \emptyset, \{\emptyset\} \}$

$P(S) \cap P(P(S))$

$\{\emptyset\} \cap \{\emptyset, \{\emptyset\}\} = \{ \emptyset \}$

$S = \{ 1, \{1\} \}$

$P(S) = \{ \emptyset, \{1\}, \{\{1\}\}, \{1, \{1\}\} \}$

$P(S) \cap S = \{ \{1\} \}$

$P(P(S)) = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \{\{\{1\}\}\}, \{\{1, \{1\}\}\}$ - - - -

$P(S) \cap P(P(S)) = \{ \emptyset, \{\{1\}\} \} \neq \{\emptyset\}$



Topic : Venn Diagram

Venn diagram is used to represent the relationship among the sets pictorially.

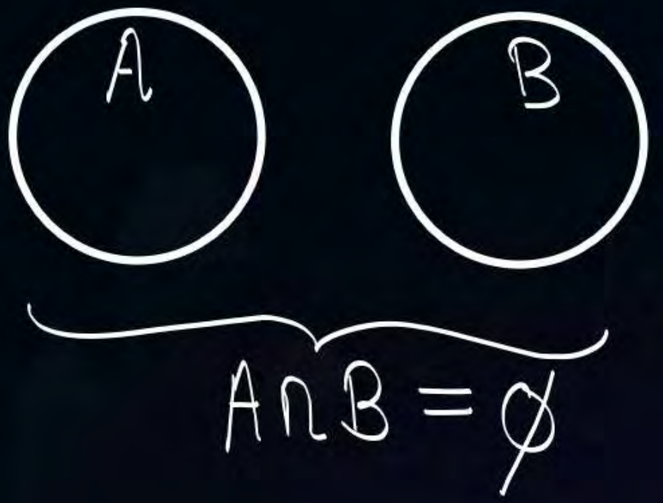


Here $B \subseteq A$.

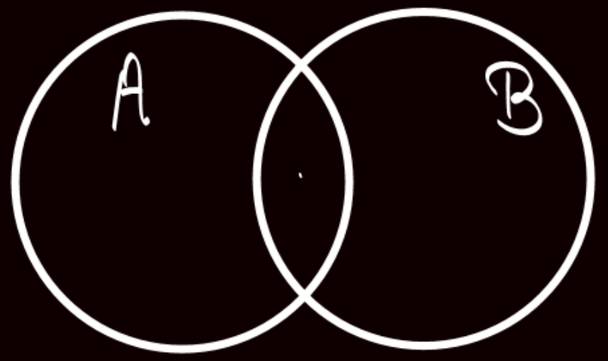
&

$B \cap A = B$

$B \cup A = A$



Slide



$A \cap B \neq \phi$



Topic : Set Operations

- Complement of a set
- Union of two sets
- Intersection of two sets
- Set difference
- Symmetric difference of two sets

Slide

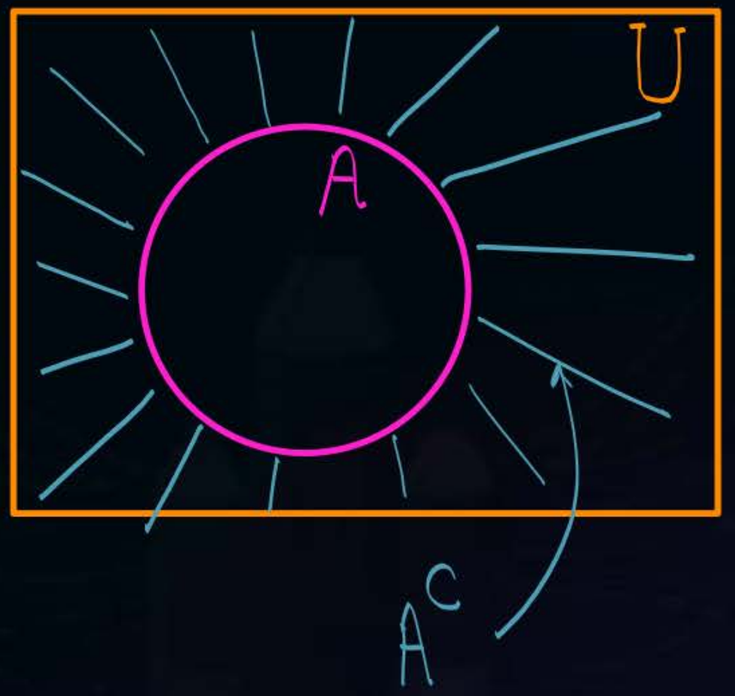


Topic : Complement of a set

Complement of a set is always defined w.r.t. universal set.

$$A^c = \{x \mid x \in U \text{ and } x \notin A\}$$

A^c → Complement of set A



$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{1, 2, 4, 7, 8, 9\}$

$A^c = \{3, 5, 6, 10\}$

Slide

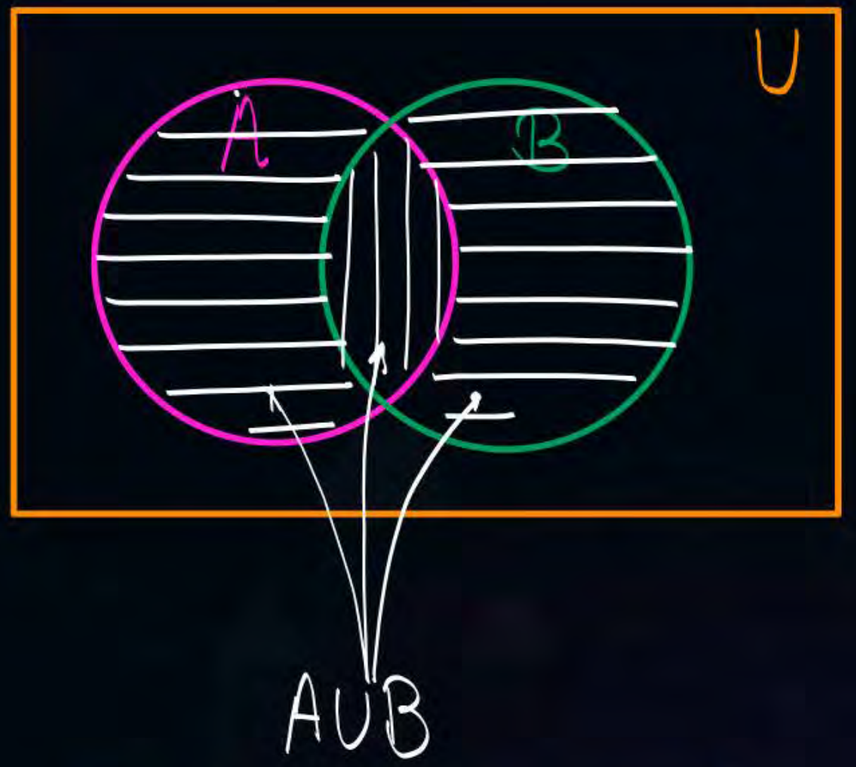


Topic : Union of two sets

Let A and B are two sets

then

$$A \cup B = \{x \mid x \in A \text{ OR } x \in B\}$$



Note: $A \cup B = B \cup A$

Slide



Topic : Intersection of two sets

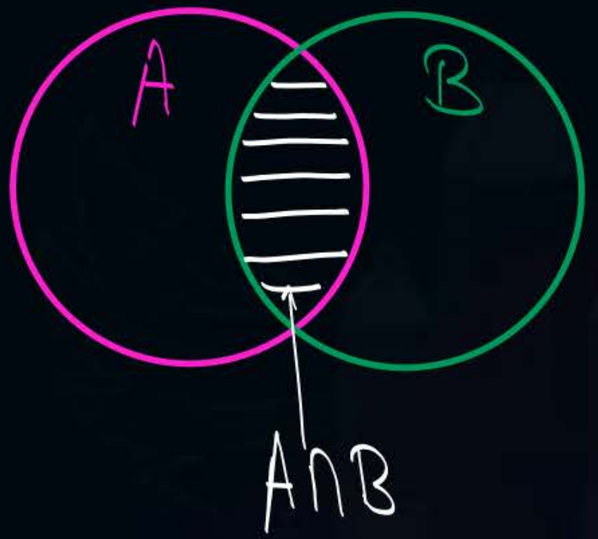
Let A & B are two sets

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$



$A \cap B = \emptyset$

{If $A \cap B = \emptyset$ then A & B are called disjoint sets}



Note: $A \cap B = B \cap A$

Slide

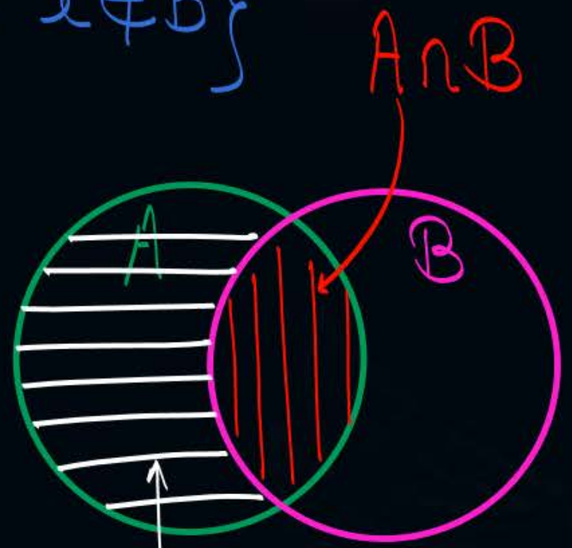


Topic : Set difference

let A & B are two sets

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

It will result all the elements that are present in set A, but not present in set B.



$$A - B = A - (A \cap B)$$

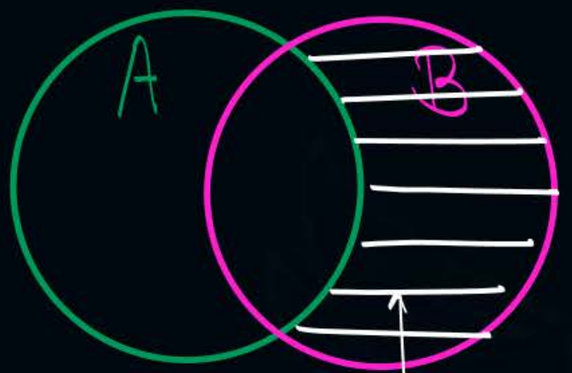
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Topic : Set difference

let A & B are two sets

$$B - A = \{x \mid x \in B \text{ and } x \notin A\}$$



$$B - A = B - (A \cap B)$$
$$= B - (B \cap A)$$

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In general,

$$A - B \neq B - A$$

$A - B = B - A$ if and only if $A = B$,

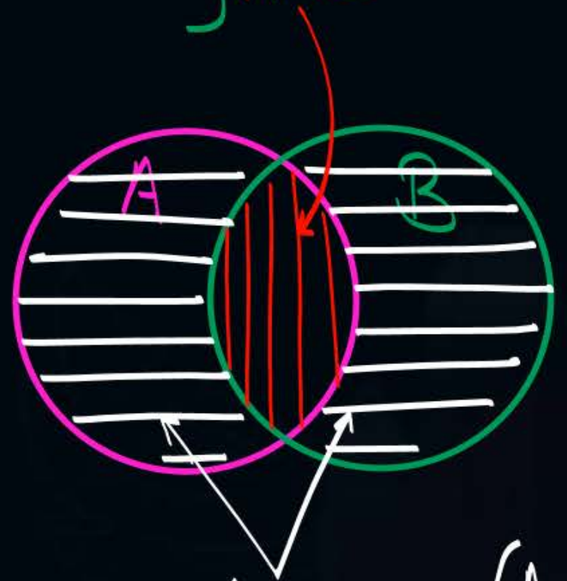
and in that case $A - B = B - A = \emptyset$



Topic : Symmetric difference of two sets

Let A & B are two sets

$$A \Delta B = \{x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin A \cap B\}$$



$$A \Delta B = (A \cup B) - (A \cap B)$$

$$A \Delta B = (A - B) \cup (B - A)$$

Note: $A \Delta B = B \Delta A$

Slide



Topic : Properties of Set Operations

1. Idempotent:

   a. $A \cap A = A$

   b. $A \cup A = A$

2. Identity:

   a. $A \cup \emptyset = A$

   b. $A \cap U = A$



Topic : Properties of Set Operations

3. Domination:

   a. $A \cap \emptyset = \emptyset$

   b. $A \cup U = U$



Topic : Properties of Set Operations

4. Complementation:

   a. $A \cup A^c = U$

   b. $A \cap A^c = \emptyset$

5. Double Complement:

   a. $(A^c)^c = A$

Slide



Topic : Properties of Set Operations

6. Commutative

   a. $A \cup B = B \cup A$

   b. $A \cap B = B \cap A$

7. Associative

   a. $A \cup (B \cup C) = (A \cup B) \cup C$

   b. $A \cap (B \cap C) = (A \cap B) \cap C$



Topic : Properties of Set Operations

8. Absorption

   a. $A \cup (A \cap B) = A$

   b. $A \cap (A \cup B) = A$

9. DeMorgan's

   a. $(A \cup B)^C = A^c \cap B^c$

   b. $(A \cap B)^C = A^c \cup B^c$



Topic : Properties of Set Operations

10. Distributive

   a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

   b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Slide



2 mins Summary

Topic Venn diagram

Topic Set operations

Topic Properties of set operations

Slide

THANK - YOU